

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

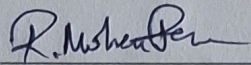
Duration: 60 Minutes

Total Marks: 25

Code of Conduct

I MOHAN RAM / 19252117 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Short Answer Test

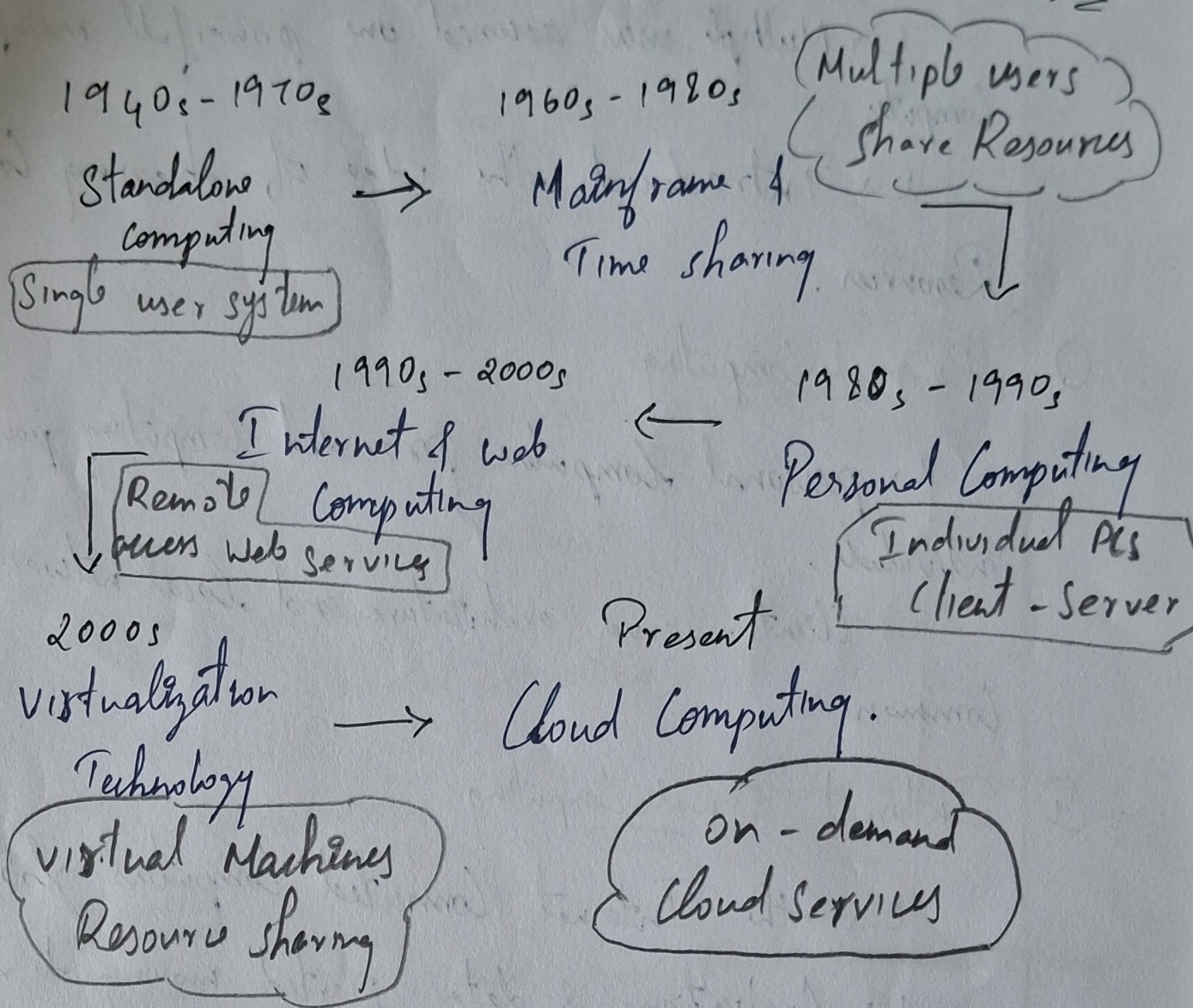
1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

①

Evolution from Standalone Computing to cloud-Based system.



Evolution Stages.

① Standalone Computing.

↳ Early Computers operated independently.

↳ A single user or organization controlled the hardware & software.

↳ Resources were expensive & limited.

② Mainframe & Time-sharing systems.

↳ Multiple users accessed one powerful mainframe computer.

↳ Introduced the idea of sharing computing resources.

③ Personal Computing

↳ personal computers provided computing power to individuals.

↳ Client-server architecture and local network became common.

④ Internet & web Computing

↳ the Internet connected computers globally.

↳ Applications & data started moving to remote servers.

⑤ Virtualization Technology.

↳ A single physical server could run multiple virtual machines.

↳ Improved resource utilization & enabled flexible scaling.

⑥ Cloud Computing.

↳ Computing resources such as storage, server, & software are provided through the internet.

↳ users can access services anytime & pay based on usage.

Technologies Enabling Cloud Computing.

→ high-speed internet.

→ virtualization.

→ Distributed Computing.

→ data centers.

→ Automation & service-oriented architecture.

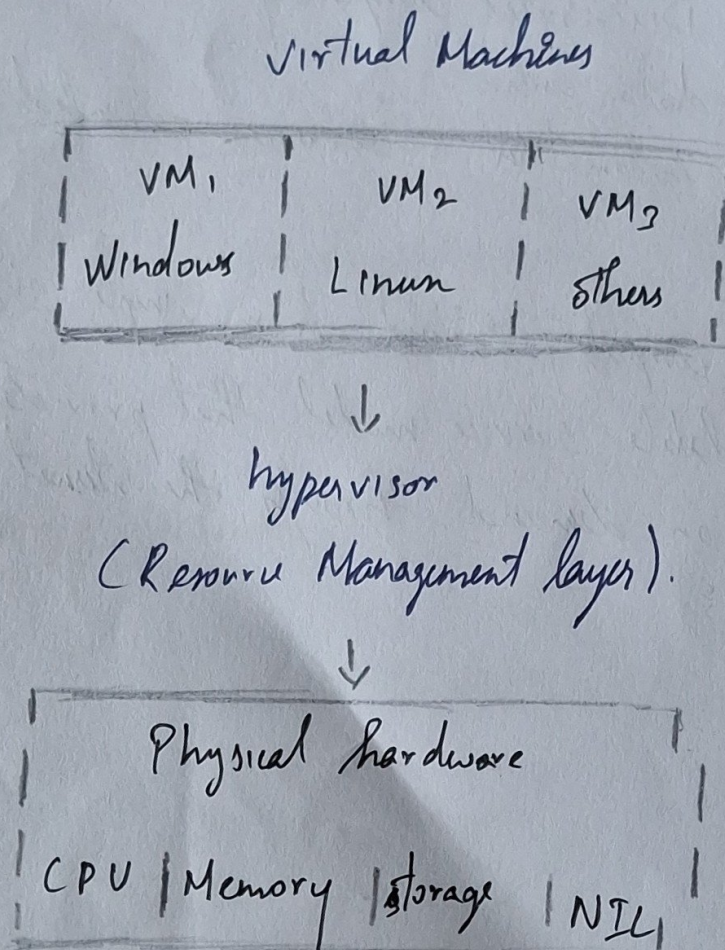
Conclusion.

→ Cloud Computing evolved from simple standalone computers into a scalable service model that provides computing resources on demand through the internet.

② Working of hypervisors in Virtualization

A hypervisor is software or firmware that creates and manages virtual machines (VMs) on a physical computer. It allows multiple operating systems to run on a single hardware system by allocating resources, such as CPU, memory, storage, and network, between VMs.

Working of hypervisor



Functions of Hypervisor

- Create & runs virtual machines.
- Allocates hardware resources to VMs.
- provides isolation b/w different different operating system.
- allows multiple OS environments to run simultaneously

Types of Hypervisors

feature	Type 1 (Bare Metal)	Type 2 (Hosted)
Installation	Runs directly on Physical hardware	Runs on top of a host operating system.
Performance	Faster and more efficient	Slower due to extra OS layer.
usage	Used in data centers & cloud environments	Used for personal & development purposes.
Security	More secure	less secure compared to type 1.
Example	VMware ESXi, Microsoft Hyper-V, Xen.	VMware Workstation, Oracle Virtual Box, parallels Desktop.

Type 1 hypervisor

VM OS



hypervisor



hardware

Type 2 hypervisor

~~VM~~ VM OS



hypervisor



host operating system



hardware.

Conclusion.

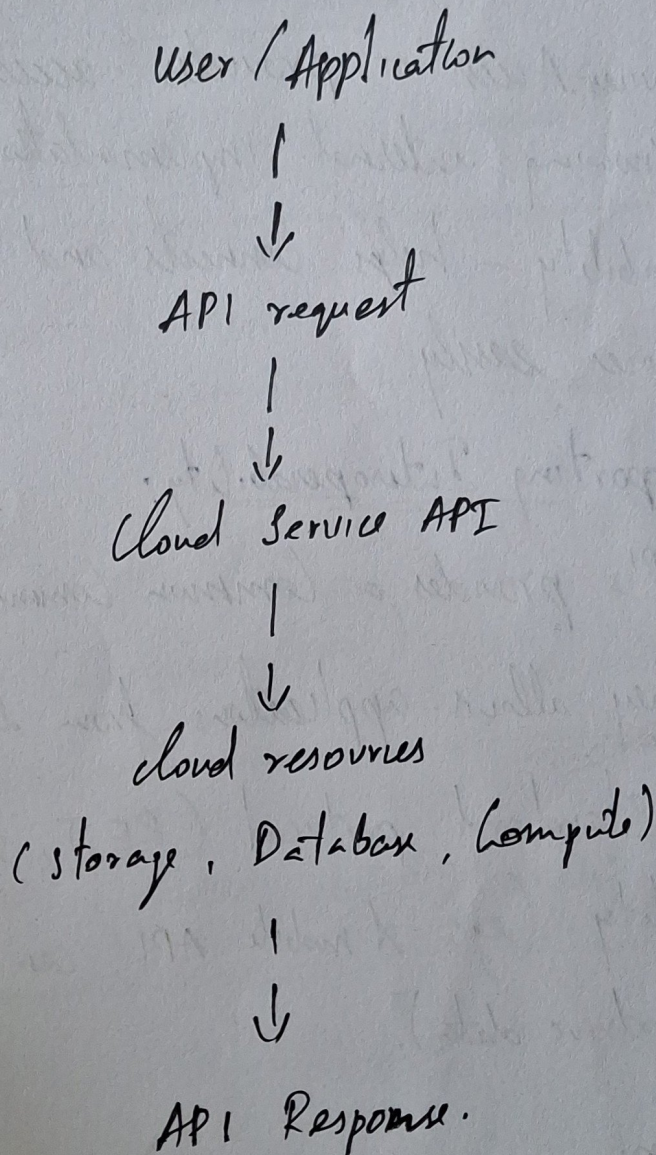
Type 1 provides better performance and are mainly used in cloud data centers, while type 2 are suitable for desktop users and testing environments.

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③ Role of APIs in cloud service integration.

API (Application Programming Interface) is a set of that allows different software applications & cloud s. To communicate and exchange data with each other.

Working of APIs in cloud:



Role of APIs in Cloud integration

→ Communications - enables applications to interact with cloud services.

→ Automation - allows automatic management of cloud resources.

→ Data exchange - Transfers data between different applications and platforms.

→ Service Access - provides access to cloud features without knowing internal implementation.

→ Scalability - helps connect and manage multiple cloud services easily.

APIs Supporting Interoperability:

→ APIs provides a Common Communication interface.

→ They allows applications from different vendors ... etc.

→ Standard protocol (REST, SOAP, JSON) improve Compatibility. (eg: A mobile API can use cloud storage to store or retrieve data).

Conclusion.

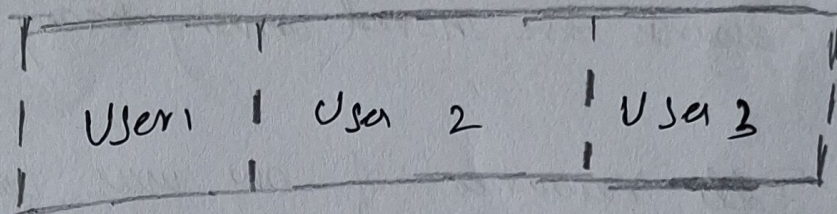
APIs act as a bridge b/w cloud services, enabling seamless integration, interoperability & efficient communication among different applications & platforms.

④ Concept of Resource pooling in cloud computing.

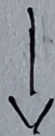
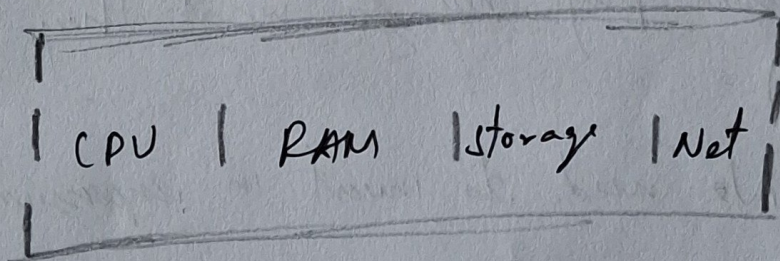
Resources pooling is a cloud computing concept where a provider combines (pools) physical & virtual resources - such as CPU, memory, storage, & network - To serve multiple users using a shared Infrastructure..

Working of Resource pooling.

Multiple Users (Tenants)



cloud Provider
(Resource pool)



Dynamic Allocation to users.

→ Resources are dynamically assigned and reassigned based on user demand.

→ users do not know the exact physical location of their resources (location independence).

→ This model is also called multi-tenancy.

Benefits of Resources Pooling for Multiple Users.

① Efficient Resources Utilization.

→ Resources are shared among many users, reducing wastages.

→ Idle resources of one user can be used by another.

② Cost Reducing.

→ users pay only for what they use (pay-as-you-go model).

→ No need to invest in expensive hardware.

③ Scalability & flexibility

→ Resources can be increased or decreased instantly based on demand.

→ Supports both small & large workloads.

④ High Availability

→ If one resource fails, others in the pool can take over.

→ Ensure continuous services for users.

⑤ Isolation & Security

→ Even though resources are shared, each user's data is kept separated & secure.

Conclusion

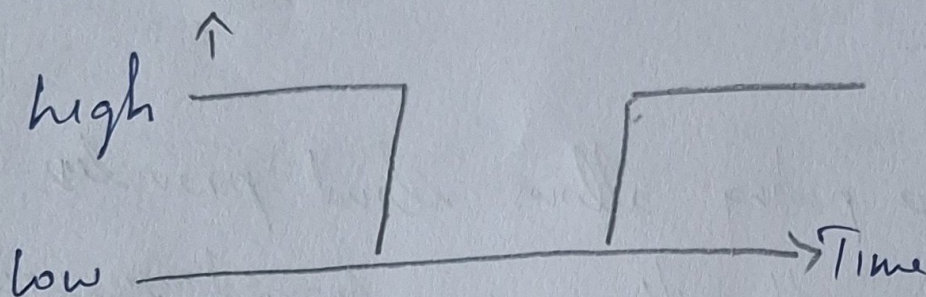
Resource pooling allows cloud providers to serve multiple users efficiently using shared resources. It improves cost efficiency, scalability, and performance, making cloud computing highly flexible and accessible.

⑤ Rapid elasticity in cloud computing.

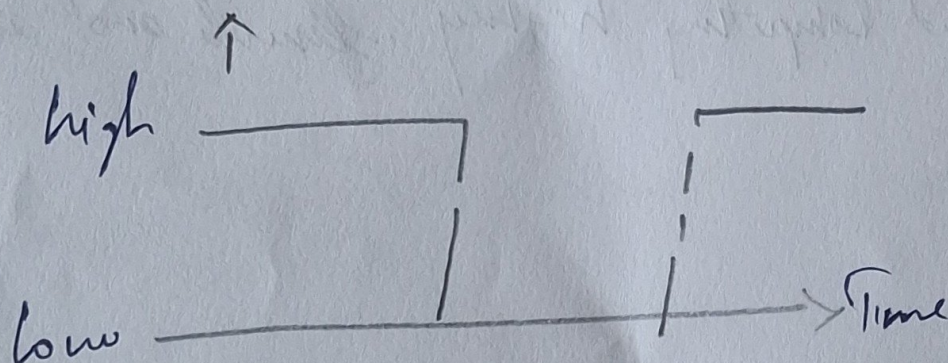
Rapid elasticity is the ability of a cloud system to quickly scale resources up or down based on demand. To users, resources appear almost unlimited and can be provisioned or released at any time.

Working of Rapid elasticity.

Workload demand.



Resources allocated.



→ when demand increases → more resources are added (Scale up)

→ when demand decreases → resources are removed (Scale down)

how it supports Dynamic Workloads

① Handles demand Fluctuations

→ Automatically adjusts resources during peak and low usage.

→ Example :: E-commerce sites during sales.

② Cost Efficiency.

→ Users pay only for the resources they use.

→ Avoids over-provisioning and wastage.

③ High performance.

→ Ensures applications run smoothly even under heavy load.

→ Prevents system slowdowns.

④ Automation.

→ Scaling is often automatic using predefined rules or policies.

→ No manual intervention needed.

⑤ Flexibility.

→ Supports unpredictable and changing workloads easily.

Conclusion

→ Rapid elasticity allows cloud system to adapt instantly to workload changes, ensuring efficient performance, scalability, and ~~cost~~ cost savings for dynamic applications.